

The Road to a Cleaner Kickoff: Finding the Most Sustainable Super Bowl Host

This study develops a quantitative and scientifically grounded framework for evaluating the environmental sustainability of major sports events, with the goal of selecting the most suitable host city for the Super Bowl in 2029. The framework integrates the **AHM-PCA-PHM hybrid** weighting method, an improved **TOPSIS** evaluation model, the **GM(1,1) grey prediction** model, grey relational analysis, and multi-dimensional sensitivity testing. Based on six environmental dimensions—**carbon emissions, primary energy, transportation emissions, water use, waste, and accommodation and catering**—a system of **24 indicators** is constructed, and two additional dimensions (Legacy Sustainability and Multi-Venue Coordination) are incorporated for broader event types.

Subjective weights are derived through the Analytical Hierarchy Model, while objective weights are determined through Principal Component Analysis. These are harmonized using the PHM method to obtain integrated weights, which are then applied in the TOPSIS framework to assess 19 legacy Super Bowl host cities. This hybrid approach ensures that expert judgment and statistical information jointly influence the final ranking.

To address the requirement of forecasting future performance, the GM(1,1) model is used to project key environmental indicators to 2029. Model validation using data from **Super Bowl LIX** confirms that the system accurately captures patterns in energy consumption, transportation demand, waste generation, and water use, demonstrating strong reliability and interpretability.

The results show that **transportation emissions (weight 0.32)** and **energy-related indicators (weight 0.26)** are the dominant drivers of overall environmental burden. Minneapolis achieves the highest sustainability score (**0.780**), benefiting from a cleaner future electricity grid, lower transportation emissions, and efficient stadium operations. Lower-ranking cities such as Pontiac, Pasadena, and Jacksonville are limited by higher projected travel emissions and less favorable energy structures. Among potential first-time hosts, **Seattle** performs best overall, with **Denver** and **Philadelphia** each showing strengths in specific dimensions.

Grey relational analysis identifies **infrastructure investment, energy mix transition, and transportation** upgrades as the most influential structural factors. Sensitivity analysis further reveals that single-venue events like the Super Bowl are most sensitive to transportation, energy, and venue operations, while multi-venue events such as the Olympics depend more on legacy outcomes and venue coordination.

Overall, the study establishes a comprehensive **sustainability assessment model**, validates its performance, forecasts future environmental conditions, and identifies key influencing factors. Based on all integrated results, **Minneapolis** is recommended as the most environmentally sustainable and future-compatible host city for the Super Bowl in 2029.

Keywords: Sustainability Assessment; Hybrid Weighting Model; Grey Prediction (GM(1,1)); TOPSIS Evaluation; Environmental Indicators; Host City Selection

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1 Introduction

1.1 Problem Background

Every year, the Super Bowl brings together over 100 million viewers and hundreds of thousands of traveling fans, making it one of the world's largest and most resource-intensive sporting events. A typical Super Bowl produces nearly 55 to 60 tons. Although organizations like the NFL and FIFA have begun addressing these issues, most of their sustainability efforts still focus on offsetting emissions after the event instead of tackling them beforehand. For instance, during the 2015 Super Bowl in Phoenix, the city aimed to divert 80% of game-day waste from landfills but ultimately reached only 72%, despite major financial and labor investments.

Thus, establishing a clear, quantitative framework for evaluating how sustainable potential host cities are has become a pressing issue. Such a model would enhance decision-making efficiency, strengthen economic sustainability, and raise public awareness about environmental responsibility while reducing the need for expensive post-event cleanup.

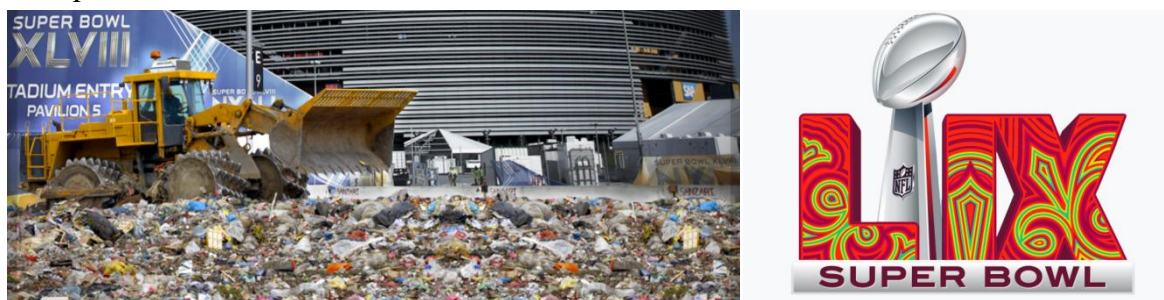


Figure 1 Massive amounts of solid waste have been generated from the Super Bowl

1.2 Restatement of the Problem

Considering the background information and goals identified in the problem statement, our team aims to address the following problems:

- (1) **Defining Sustainability Metrics:** Identify the key environmental indicators that determine the sustainability performance of the Super Bowl.
- (2) **Model Validation:** Evaluate the proposed model using data from Super Bowl LVIII.
- (3) **Optimal Host Selection:** Apply the model to determine which U.S. city would be the most environmentally sustainable host for the Super Bowl in 2029.
- (4) **Sensitivity Assessment:** Conduct a sensitivity analysis to examine how changes in various factors may influence each city's sustainability ranking.
- (5) **Non-Technical Communication:** Prepare a non-technical summary for the NFL Green Committee that introduces the model and explains our findings in clear, accessible language.

1.3 Literature Review

Previous sustainability research on mega-sporting events has largely relied on post-event assessments of emissions and waste. Studies show that transportation and energy use are the dominant carbon sources, as seen in the London 2012 Olympics^[1] and the FIFA World Cup^[2].

To systematize evaluations, scholars have introduced quantitative multi-criteria methods such as AHP, entropy weighting, TOPSIS, and LCA. For example, Tang ^[3] applied an AHP-TOPSIS model to urban sustainability and demonstrated that weighted indicators (e.g., GHG emissions, transport, waste) improve comparability. Wang ^[4] used LCA for the Beijing 2022 Winter Olympics and found that fan travel and venue energy accounted for over 70% of emissions. A broader review by Cerezo-Estevé et al. ^[5] noted that few frameworks assess sustainability before events occur. Addressing this gap, our project develops a predictive AHP-TOPSIS model to rank potential Super Bowl host cities, shifting sustainability planning from reactive post-event analysis to proactive decision-making.

1.4 Our Work

The topic requires us to establish a model to evaluate the environmental sustainability of large-scale Super Bowl host cities according to multiple environmental and operational indicators. The model is then applied to assess the validity of past Super Bowl host selections, predict future suitability, and undergo sensitivity testing. Our work mainly includes the following steps.

- ✧ **Model Construction:** Develop an environmental sustainability indicator system; Build a TOPSIS-based model to evaluate Super Bowl host cities.
- ✧ **Application to Super Bowl Cities:** Apply the model to 19 past host cities; Use GM(1,1) for trend prediction and complete missing data for potential host cities; Evaluate the three potential cities for future hosting suitability.
- ✧ **Model Extension to the Olympics:** Add two long-term impact indicators to adapt the model for Olympic host evaluation; Use grey relational analysis for indicator weighting and recommend suitable Olympic host cities.
- ✧ **Strategies and Sensitivity Analysis:** Propose sustainability improvement strategies; Conduct sensitivity analysis to test model robustness and identify key influencing indicators.

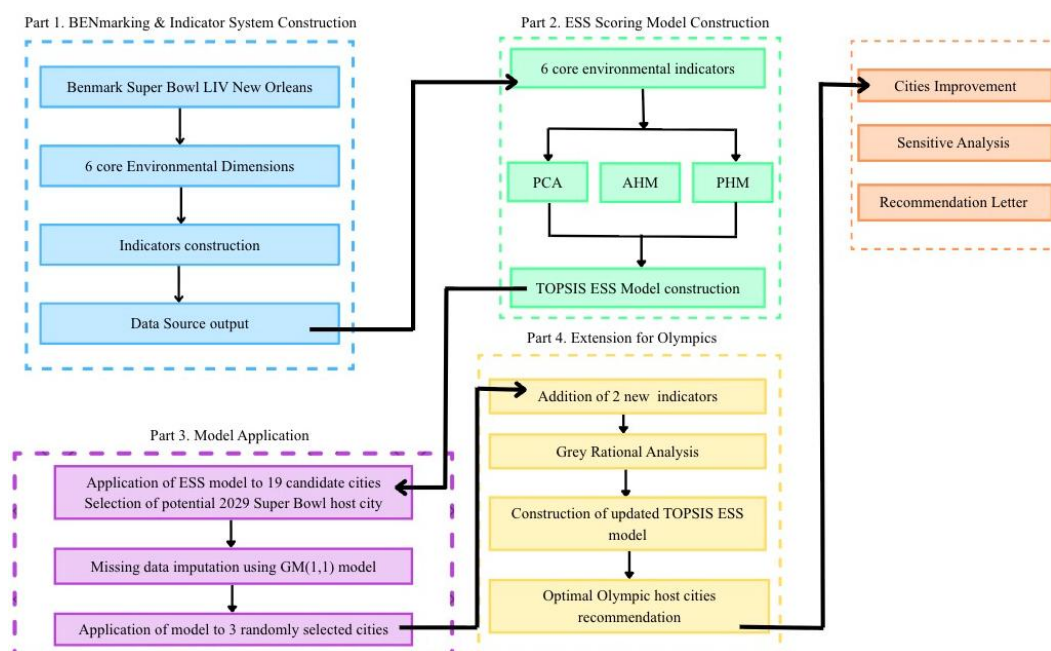


Figure 2 Model Overview

2 Assumptions and Justifications

- **Assumption 1:** The accuracy of historical data is ensured

Justification: Historical datasets used for calibration and validation are assumed to be accurate and reliable, providing stable parameter estimation and supporting valid comparisons across years.

- **Assumption 2:** Spectator travel dominates the total environmental footprint

Justification: Spectator travel is widely recognized as the largest contributor to emissions for large, one-time events. Thus, the model focuses on air, car, and local transportation while omitting smaller contributors such as venue construction and broadcasting.

- **Assumption 3:** Transportation modes operate under constant occupancy rates

Justification: Standard occupancy levels (e.g., 1–2 persons per car, average airline load factors) are used to calculate per-passenger-kilometer emissions. This avoids uncertain event-specific data and allows consistent comparison across cities.

- **Assumption 4:** Travel mode choice depends solely on distance

Justification: The choice between car and air travel is modeled using a logistic function based only on travel distance, a reliable and universally available predictor that ensures consistent estimation across host cities.

3 Notations

The key mathematical variables used in this paper are listed in Table 1.

Table 1 Notations

Symbol	Definition	Unit
T_{car}	Total per-capita travel distance of non-local audiences by car	km/person
T_{air}	Total per-capita travel distance of non-local audiences by air	km/person
T_{local}	Average round-trip travel distance for local audiences	km
T_{density}	Average residential density around the venue	persons/km ²
T_{road}	Proportion of road investment to GDP	%
T_{public}	Proportion of public transport investment to GDP	%
S_{water}	Falkenmark water stress index	
S_{energy}	Economic energy pressure index for the city	
R_{clean}	Proportion of renewable energy in the local grid	%
R_{ev}	Proportion of electric vehicles in the civilian passenger fleet	%
R_{recycle}	Mass proportion of municipal solid waste that is recycled	%
R_{reuse}	Proportion of treated wastewater directly reused	%
P_{noise}	Increase in noise level around the venue during the event	dB(A)
P_{light}	Relative increase in nighttime radiance around the venue	Dimensionless

4 Super Bowl Indicator System and Model Establishment

4.1 Evaluation Indicators

4.1.1 In-Event Carbon Emissions (E)

The rationale for establishing this indicator is supported by both policy mandates and heightened public awareness. Global climate frameworks—particularly the Paris Agreement's commitment to limiting warming to 1.5 °C—impose stringent requirements that make carbon emission mitigation a necessary objective for large-scale sporting events. Concurrently, the escalating severity of climate change has intensified public scrutiny, underscoring the importance of accurately assessing and reducing event-related emissions. The carbon emission formula is given by:

$$E = E_{\text{elec}} + E_{\text{fuel}}$$

We decompose carbon emissions into three components: electricity emissions, direct combustion emissions, and waste-related emissions.

✧ Electricity Emissions E_{elec}

This indicator represents the venue's ability to maintain high performance over time while minimizing environmental burden, reflecting its efficiency, adaptability, and equity across different regions, technologies, and climate conditions. The formula is:

$$E_{\text{elec}} = D_{\text{elec}} \times EF_{\text{grid}}(\text{city, month})$$

In this equation, E_{elec} represents the total carbon emissions from electricity consumption, measured in tonnes of carbon dioxide equivalent (tCO₂e), D_{elec} represents the electricity consumption, measured in kilowatt-hours(kWh).

$EF_{\text{grid}}(\text{city, month})$ represents the local grid's average emission factor, measured in tonnes of carbon dioxide equivalent per kilowatt-hour (tCO₂e/kWh). This factor is incorporated because the primary sources of power generation can vary by city and time (e.g., greater reliance on clean energy in summer, coal in winter), significantly affecting carbon emissions.

✧ Direct Combustion Emissions: E_{fuel}

This indicator involves the systematic accounting of all on-site combustion sources (e.g., boilers, backup generators, and other fossil fuel-driven equipment) to ensure comprehensive coverage of Scope 1 emissions. The formula is:

$$E_{\text{fuel}} = \sum_j Q_j \cdot EF_j$$

E_{fuel} represents the carbon emissions from direct fuel combustion, measured in tonnes of carbon dioxide equivalent (tCO₂e), Q_j represents the consumption volume of the j-th fuel type, measured in cubic meters (m³), EF_j represents the emission factor for the j-th fuel type, measured in tonnes of carbon dioxide equivalent per cubic meter (tCO₂e/m³).

4.1.2 Total Primary Energy Consumption (Primary Energy)

This indicator is the total primary energy equivalent of electricity and fuel consumption during the event. It provides an energy intensity metric for comparing the energy efficiency

of different events. Its mathematical expression is:

$$\text{Primary Energy} = \text{Electricity Energy} + \text{Fuel Energy}$$

PrimaryEnergy is the total primary energy consumption, measured in Megawatt-hours(MWh).

✧ **Stadium's Primary Electricity energy consumption(Electricity Energy)**

$$\text{Electricity Energy} = \text{Elec_MWh} \cdot \phi_{\text{elec}}(\text{grid})$$

Electricity Energy is the primary energy consumption from electricity.Elec_MWh is the electricity consumption in MWh (MWh), $\phi_{\text{elec}}(\text{grid})$ is the grid's primary energy conversion factor.

✧ **Stadium's Primary Fuel energy consumption(Fuel Energy)**

$$\text{FuelEnergy} = \sum_f Q_f \cdot \phi_f$$

Fuel Energy is the primary energy consumption from fuel. Q_f is the consumption volume of the f-th fuel, measured in cubic meters(m^3), ϕ_f is the calorific value conversion factor for the f-th fuel, measured in megawatts per cubic mete (MWh/m^3).

4.1.3 Transportation Emissions(E_{trans})

This indicator represents the emissions generated from commuting during a large-scale event. Its mathematical expression is:

$$E_{\text{trans}} = \sum_{\text{mode}} \text{PaxKilos}_{\text{mode}} \cdot \text{EF}_{\text{mode}}(\text{region})$$

E_{trans} represents the total emissions from transportation, measured in tonnes of carbon dioxide equivalent), $\text{PaxKilos}_{\text{group}}$ represents the passenger miles traveled by certain ways

(e.g., air, road) . Its mathematical expression is:

$$\text{PaxKilos}_{\text{mode}} = \text{people} \cdot \text{average distances} \cdot \text{transportation mode}$$

$\text{EF}_{\text{mode}}(\text{region})$ represents the regionalized emission factor for a specific transportation mode, measured in tonnes of carbon dioxide equivalent per person-kilometer

($\text{tCO}_2\text{e}/(\text{person} \cdot \text{kilometers})$)The 'mode' includes various commuting methods such as taxis, airplanes, and public transport.

4.1.4 Water Resource(Water)

This indicator is the total freshwater withdrawal and wastewater discharge during the event. It is used to assess water resource stress, particularly for events in arid regions. (Water) represents the volume of water usage, measured in cubic meters

4.1.5 Total Waste W_{total}

This indicator is the total mass of waste handled, used to assess the total solid waste generated by the event and its disposal pathways. W_{total} represents the total weight of waste, measured in tonnes (ton).

4.1.6 Accommodation and Catering

This category encompasses carbon emissions from all accommodation and catering activities.

✧ Accommodation Emissions (E_{lodging})

$$E_{\text{lodging}} = \sum_c \text{Roomnights}_c \cdot EF_{\text{hotel}}(\text{city, class})$$

E_{lodging} represents the total carbon emissions from accommodation, measured in tonnes of carbon dioxide equivalent (tCO_2e). C represents hotel class. Roomnights_c represents the number of room-nights in distance zone ($\text{Room} \cdot \text{nights}$). The mathematics expression is

$$\text{Roomnights} = \sum_i N_{\text{people},i} \cdot \text{Days}_{\text{stay},i} \cdot \text{adjustment coefficient}_i$$

Where $N_{\text{people},i}$ is the number of people in category. $\text{Days}_{\text{stay},i}$ is their length of stay, and adjustment coefficient i is the probability of this category staying in hotels.

$EF_{\text{hotel}}(\text{city, class})$ is the hotel emission factor, representing the carbon intensity per room-night ($\text{tCO}_2\text{e}/(\text{Room} \cdot \text{nights})$). Its value is influenced by hotel category, the cleanliness of the city's grid, and season.

✧ Food service Emissions E_{foods}

This indicator is a significant part of the carbon footprint. Catering emissions are strongly correlated with dietary structure; increasing the proportion of vegetarian meals can significantly reduce the carbon footprint. The mathematical expression is

$$E_{\text{foods}} = \sum_k \text{Servings}_k \cdot EF_{\text{meal},k} \cdot (1 - \text{veg-share uplift})$$

E_{foods} represents the total carbon emissions from food service, measured in tonnes of carbon dioxide equivalent (tCO_2e). Servings_k represents the total number of servings for each food type k provided during the event, measured in (Servings). Food types are grouped based on differences in carbon footprint. $EF_{\text{meal},k}$ represents the full life-cycle carbon footprint per unit mass of the food item, measured in kilograms of carbon dioxide equivalent per kilogram ($\text{kgCO}_2\text{e}/\text{kg}$). The magnitude of the emission factor is determined by the type of food. $(1 - \text{veg-share uplift})$ is the vegetarian intervention parameter, representing the emission reduction coefficient achieved by increasing the proportion of plant-based foods. Its value represents the share of vegetarian meals.

According to above, we incorporated seven 24 indicators from 6 dimensions to quantify the environmental impact of a major sports event, as shown in **Figure 3**.

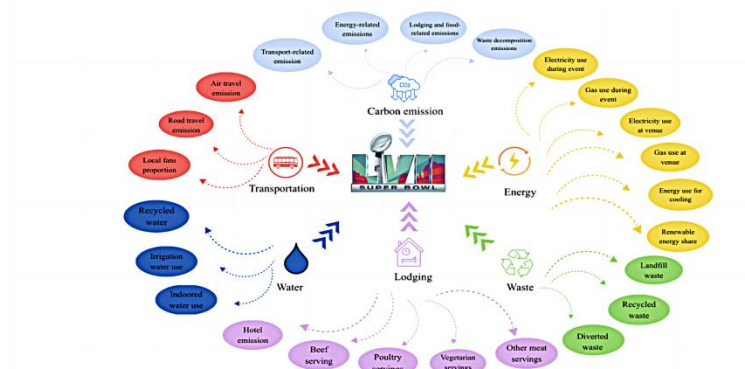


Figure 3 The 6-dimensional Indicators

4.2 Establishment of the evaluation model

In this section, we evaluate the environmental sustainability of potential Super Bowl host locations by constructing an AHM-PCA-PHM integrated model. This model focuses on six key environmental indicators, including carbon emissions, energy consumption, transportation, water usage, waste generation, and lodging/food systems, which integrates both subjective and objective weighting approaches. The AHM method is applied to capture expert judgment on the relative environmental importance of each indicator, while the PCA method extracts the weights based on the statistical variance among indicators.

Building on these, the PHM method is introduced to harmonize AHM and PCA weight results, projecting them onto a decision axis to achieve consistency and equilibrium in the final determination. This combination ensures a balanced and credible evaluation for both professional insights and experience evidence. To quantify the sustainability performance of each city, an improved TOPSIS is developed to calculate an environmental sustainability score. A process flowchart of the model is shown in Fig. 4, providing a clear visualization of this methodology. Through this integrated model, we aim to establish a scientific and quantifiable methodology for evaluating the environmental suitability for Super Bowls host cities, contributing to greener and more sustainable future event planning. A process flowchart for the model is given in **Figure 4**:

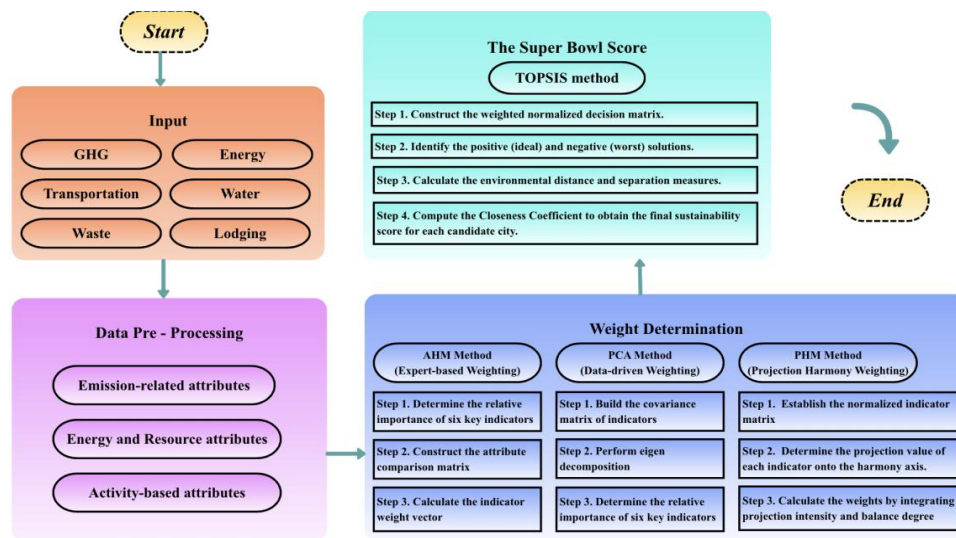


Figure 4 Olympic SDE Scoring Framework

4.2.1 Integrated Weight Determination (AHM–PCA–PHM)

Assume there are n evaluation factors, and m alternatives (candidate host cities). We first obtain subjective weights using an Analytical Hierarchy Model (AHM), then derive objective weights from Principal Component Analysis (PCA), and finally combine them into a set of hybrid PHM weights.

✧ AHM-based subjective weights

Step1: Pairwise comparison matrix

Experts compare every pair of factors using a 1 – 9 preference scale. The comparisons form a judgment matrix

$$B = (b_{ij})_{n \times n}$$

Where expresses b_{ij} the importance of factor C_i relative to factor C_j . The matrix satisfies

$$b_{ij} > 0, b_{ji} = \frac{1}{b_{ij}}, b_{ij} = 1.$$

Step2: Preference transformation

Let s_{ij} in $\{0,1,\dots,8\}$ be the level difference between C_i and C_j on the 1 - 9 scale (0 means “equally important”, 8 means “extremely more important”). Then we define

$$p_{ij} = f(x) = \begin{cases} \frac{2s_{ij}}{2s_{ij} + 1}, i \neq j, b_{ij} > 1 \\ \frac{1}{2s_{ij} + 1}, i \neq j, b_{ij} < 1 \\ 0.5, i \neq j, b_{ij} = 1 \\ 0, i = j \end{cases}$$

Step3: AHM weights

For factor C_k , its initial subjective weight is defined as the average preference of C_k over all other factors:

$$\omega_k^{(A)} = \frac{1}{n-1} \sum_{\substack{j=1 \\ j \neq k}}^n p_{kj}, k = 1, 2, \dots, n.$$

To make the weights sum exactly to 1, we renormalize:

$$\tilde{\omega}_k^{(A)} = \frac{\omega_k^{(A)}}{\sum_{r=1}^n \omega_r^{(A)}}, k = 1, 2, \dots, n.$$

The vector $\tilde{\omega}^{(A)} = \{\omega_1^{(A)}, \omega_2^{(A)}, \dots, \omega_n^{(A)}\}$ is the AHM subjective weight vector.

✧ PCA-based objective weights

Now we infer objective weights directly from the data. Let the standardized data matrix be

$$Y = (y_{ik})_{m \times n},$$

where y_{ik} is the standardized value of factor C_k for alternative i .

Step1: Covariance matrix

$$O = \frac{1}{m-1} Y^T Y.$$

Step2: Eigenvalue decomposition

We compute the eigenvalues λ_t and eigenvectors e_t of H from:

$$Oe_t = \lambda_t e_t, t = 1, 2, \dots, n.$$

Let λ_1 be the largest eigenvalue and $e_1 = (e_{11}, e_{12}, \dots, e_{1n})$ its corresponding eigenvector.

Step3: PCA weights

The objective weight of factor C_k is obtained by normalizing the absolute entries of e_1 :

$$\omega_k^{(p)} = \frac{|e_{1k}|}{\sum_{r=1}^n |e_{1r}|}, k = 1, 2, \dots, n.$$

✧ PHM hybrid weights

To incorporate both expert judgement and data-driven information, we build a PHM (AHM–PCA Hybrid Method) weight for each factor.

Step1: Geometric coupling

For factor C_k , define an intermediate coupled weight:

$$u_k = \sqrt{\tilde{\omega}_k^{(A)} \omega_k^{(p)}}, k = 1, 2, \dots, n.$$

Step2: Final normalization

The final hybrid weight of C_k is

$$\omega_k^{(H)} = \frac{u_k}{\sum_{r=1}^n u_r}, k = 1, 2, \dots, n.$$

The vector $w^{(H)} = (w_1^{(H)}, \dots, w_n^{(H)})$ is used as the final weight vector in the ITOPSIS evaluation and satisfies

$$\omega_k^{(H)} \geq 0, \sum_{k=1}^n \omega_k^{(H)} = 1$$

4.2.2 TOPSIS Evaluation Procedure

Using the PHM weights, we adopt an improved TOPSIS-type method (ITOPSIS) to rank all alternatives. Let the original decision matrix be:

$$X = (x_{ik})_{m \times n}$$

where x_{ik} is the value of factor C_k for alternative i . Let $w^{(H)}$ be the weight vector obtained in Section 4.2.3.

✧ Normalization and weighting

(1) Normalization

For each factor C_k , we first remove units by vector normalization. If C_k is a benefit factor (larger is better), define

$$z_{ik} = \frac{x_{ik}}{\sqrt{\sum_{i=1}^m x_{ik}^2}}, k \in \text{benefit factors}$$

If C_k is a cost factor (smaller is better), define

$$z_{ik} = \frac{1/x_{ik}}{\sqrt{\sum_{i=1}^m (1/x_{ik})^2}}, k \in \text{cost factors}$$

This yields the normalized matrix $Z = (z_{ik})_{m \times n}$.

(2) Weighted normalized matrix

We then apply the PHM weights:

$$y_{ik} = z_{ik} \omega_k^{(H)}, i = 1, \dots, m, k = 1, \dots, n.$$

Let $Y = (y_{ik})$ be the weighted normalized decision matrix.

✧ **Ideal and anti-ideal points**

For each factor C_k , define the positive ideal value and negative ideal value as

$$y_k^+ = \max_{1 \leq i \leq m} y_{ik}, \quad y_k^- = \min_{1 \leq i \leq m} y_{ik}$$

Collect them into vectors

$$Y^+ = (y_1^+, \dots, y_n^+), \quad Y^- = (y_1^-, \dots, y_n^-)$$

✧ **Distance to the ideal solutions**

For each alternative i , we compute its Euclidean distance to the positive and negative ideal points:

$$D_i^+ = \sqrt{\sum_{k=1}^n (y_{ik} - y_k^+)^2},$$

$$D_i^- = \sqrt{\sum_{k=1}^n (y_{ik} - y_k^-)^2}.$$

Here D_i^+ measures how far alternative i is from the best possible performance, while D_i^- measures how far it is from the worst.

✧ **Closeness coefficient and ranking**

Finally, the closeness coefficient of alternative i to the ideal solution is defined by

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}, \quad i = 1, 2, \dots, m.$$

Since $0 \leq C_i \leq 1$, a larger C_i indicates that alternative i is closer to the positive ideal and farther away from the negative ideal, and therefore more desirable. By sorting all C_i in descending order, we obtain the final ranking of the alternatives.

4.3 Super Bowl suitability evaluation: GM(1,1) Prediction Model

In order to forecast the long-term environmental sustainability performance of candidate host cities, we adopt the GM(1,1) Grey Prediction Model, a classical forecasting method applicable to systems with small samples, incomplete information, and clear monotonic trends. Given the limited availability of long-term historical environmental indicators for Super Bowl host cities, GM(1,1) provides a suitable tool for generating reliable trend projections.

Step 1: Accumulated Generating Operation (AGO)

The Accumulated Generating generation is a method of successively adding data in the same sequence to generate new data. The accumulated sequence becomes the cumulative generated sequence. This is because in the gray prediction model, first-order differential equations are used for modeling, and the solution of first-order differential equations usually requires that the original sequence is generated cumulatively.

The Original Indicator's Data Sequence form would be

$$X^{(0)} = \{x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(n)\}$$

In order to reduce randomness, we perform the first order AGO. The mathematics form of the Sequence is

$$X^{(1)} = \{x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(n)\}$$

with

$$x^{(1)}(t) = \sum_{i=1}^t x^{(0)}(i), t = 1, 2, \dots, 2.$$

This transformation creates a increasing data sequences that reflect the development trend of the system we created.

Step 2: Calculating Immediate Mean Sequence

The Immediate Mean Sequence is performed on the accumulated generating sequence. This process involves constructing a new data series by calculating the average values of adjacent entries from an equidistant time sequence. The mathematics expression is

$$z^{(1)}(t) = \frac{1}{2} [x^{(1)}(t) + x^{(1)}(t-1)], t = 2, \dots, n$$

Step 3: Establish GM(1, 1) Differential Equation

Generate the first-order linear differential equation. The mathematics expression is

$$\frac{dx^{(1)}(t)}{dt} + ax^{(1)}(t) = b$$

Where a represents development coefficients determining trend direction and speed, b represents the control coefficient representing an external driving factor.

Step 4: Construct Data matrix B and Data vector Y

The mathematics expression of matrix B is

$$B = \begin{bmatrix} -z(2) & 1 \\ -z(3) & 1 \\ \vdots & \vdots \\ -z(n) & 1 \end{bmatrix}$$

The mathematics expression of vector Y is

$$Y = \begin{pmatrix} x^{(0)}(2) \\ x^{(0)}(3) \\ \vdots \\ x^{(0)}(n) \end{pmatrix}$$

Thus, the least-square estimation is

$$\begin{bmatrix} a \\ b \end{bmatrix} = (B^T B)^{-1} B^T Y$$

Step 5: Complete Prediction Model

Solving the differential equation get

$$\hat{x}^{(1)}(t+1) = (x^{(0)}(1) - \frac{a}{b})e^{-ak} + \frac{a}{b}$$

Step 6: Restore predicted value

Restore the prediction data to the original sequence. The mathematics expression is

$$\hat{x}^{(0)}(t+1) = \hat{x}^{(1)}(t+1) - \hat{x}^{(0)}(t)$$

5 Super Bowl LIX sustainability assessment

5.1 Relevant parameters

The parameters form a coherent framework for assessing the sustainability of sporting events by linking infrastructure conditions, spectator travel behavior, and environmental impact factors. Transport-related coefficients—such as distance decay and the 500-km critical threshold—shape realistic mobility patterns that drive carbon and energy estimates, while standardized emission, energy-intensity, water-use, and waste values enable consistent quantification across key activity categories. Noise and light-pollution weights, together with residential density effects, capture localized environmental burdens. Overall, the system provides a concise yet comprehensive basis for evaluating the multidimensional impacts of large sporting events.

Table 2 relevant parameters

Category	Constant	Symbol	Typical Value	Notes / Source
General Factors	Road infrastructure	G_{road}	10	Marginal benefit of road improvements for emissions & noise
	factor			
	Public transport factor	G_{public}	12	Improvement effect on local emissions
Transport Choice	Residential density baseline	G_{base}	5000 ppl/km ²	Reference for noise calculations
	Distance decay	k_c	0.1	Controls drop in car travel probability with distance
	Critical distance	d_c	500 km	Where car vs. air probability is equal
Carbon Emissions	Local travel distribution	$f_{\text{distribution}}$	0.7	Adjusts for uneven local audience distribution
	Air travel	C_{air}	0.15kg CO ₂ e/(p · km)	ICAO/ICEC typical routes
	Car	C_{car}	0.25kg CO ₂ e/(p · km)	Avg. 1-person car, EPA
	Hotel	C_{hotel}	20kg CO ₂ e/(p · day)	Industry average
Energy Intensity	Food	C_{food}	5kg CO ₂ e/(p · day)	Average diet
	Air travel	E_{air}	2.5MJ/(p · km)	IATA/ICAO data
	Car (fuel)	E_{car}	2.0MJ/(p · km)	Fuel-based
	Car (EV)	E_{ev}	0.15kWh/(p · km)	Typical EV
	Venue	E_{venue}	0.5kWh/(m ² · day)	Commercial building
	Hotel	E_{hotel}	16kWh/(p · day)	Industry KPI
Air Pollutants	Food	E_{food}	8MJ/(p · day)	Supply chain energy
	Air PM2.5	A_{air}	0.01g/(p · km)	EASA estimates
	Car PM2.5	A_{car}	0.02g/(p · km)	Tailpipe + non-tailpipe
Water Use	Personal	W_{personal}	150L/(p · day)	Urban baseline

	Irrigation	W_{irrig}	1.5L/(m ² · day)	Light urban irrigation
Waste	Solid waste	M_{waste}	1.5kg/(p · day)	Large events
Noise & Light	Noise weight	α	0.7	Weight in total pollution index
	Light weight	β	0.3	Weight in total pollution index
	Residential density effect	k_{density}	0.5	Amplifies noise exposure effect

5.2 Super Bowl LIX Environmental indicators

5.2.1 Food, Lodging, and Energy-Related Emissions

The Food & Lodging concentric diagram shows that hotel room-nights and rental accommodations dominate the category's environmental impact, with the largest radial areas reflecting far greater energy and water use than food services. Smaller segments—such as food recovery and vegetarian options—still provide meaningful benefits, indicating that targeted meal-planning and surplus-food programs can effectively reduce waste and emissions. The Energy & GHG bar chart shows a clear imbalance: electricity emissions (6.1t CO₂) exceed natural-gas emissions (≈4t CO₂), totaling about 10t CO₂. Electricity accounts for roughly 1.5 times the gas component, underscoring stadiums' heavy reliance on grid power. Increasing renewable-energy sourcing or deploying on-site clean energy could therefore deliver substantial emission reductions. Related results are shown in Figure 5.

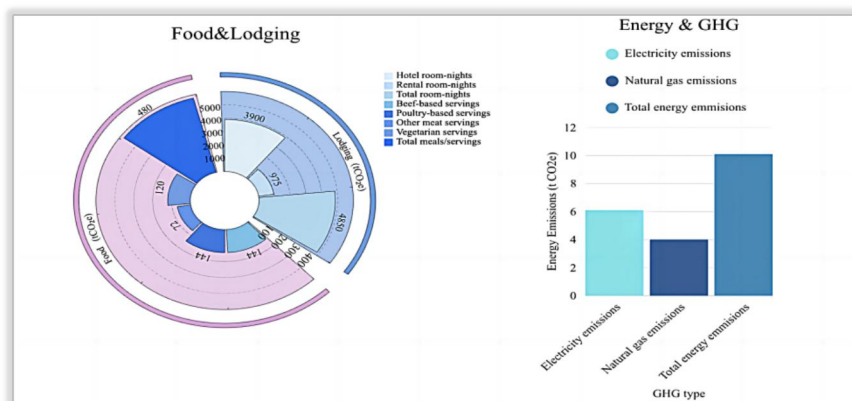


Figure 5 Environmental Burden of Food, Lodging, and Energy-Driven GHG Emissions

5.2.2 Waste Diversion Performance and Water Consumption Patterns

The Waste chart shows that about 250 units were recovered and only around 20 units were landfilled, for a total of roughly 270 units. This yields a recovery rate above 90%, indicating an effective diversion system. However, the overall waste volume is still high, highlighting the need for upstream waste-reduction measures in addition to strong recycling performance. The Water Use chart indicates that indoor consumption (1,010 m³) dominates total demand, far exceeding outdoor uses such as landscape irrigation (350 m³) and field irrigation (150 m³). Total water use reaches about 1,200 m³, suggesting that the main reduction potential lies in improving indoor efficiency (e.g., low-flow fixtures, greywater reuse), while better irrigation management offers secondary benefits. Solid waste and water-use distributions are illustrated in Figure 6.

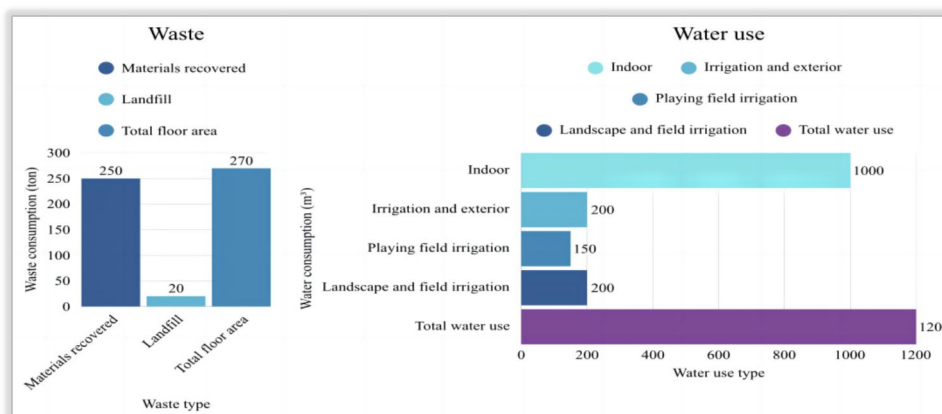


Figure 6 Solid Waste Outcomes and Water Use Distribution

5.2.3 Transportation Energy Demand and Stadium Energy Profile

The Transportation results show a sharp contrast between air and local travel. Air-travel energy reaches several million units, while local travel contributes only a few hundred thousand, meaning total transportation demand is essentially driven by aviation. This highlights that long-distance spectator and team travel—not city mobility systems—is the primary transportation impact, emphasizing the importance of geographic accessibility when selecting host cities. The Energy breakdown for stadium operations shows just over 100 units distributed across lighting, concourse areas, and back-of-house systems, with electricity as the dominant carrier and natural gas/district energy contributing far less. Compared with transportation, operational energy is a secondary impact, reinforcing that event location and associated travel demand are the main drivers of overall environmental performance. Results are shown in Figure 7.

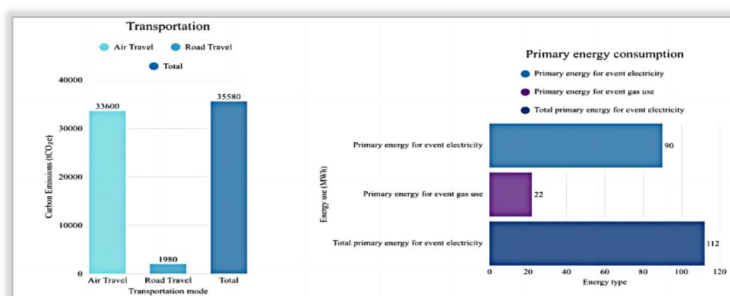


Figure 7 Transportation-Related Energy & Stadium Operational Energy Structure

5.3 Weight result

The weight results reveal a clear hierarchy among environmental factors. Transportation holds the highest weight (0.32), showing that travel—especially air travel—is the dominant source of impact. Energy and GHG emissions follow (0.26), underscoring the influence of electricity and natural gas in stadium operations. Together, these categories account for more than half of the total weight, highlighting the importance of geographic accessibility, travel volume, and regional grid carbon intensity. The remaining factors—Food & lodging (0.18), Waste (0.14), and Water (0.10)—form a secondary tier, reflecting their comparatively smaller contribution and the efficiencies already present in waste recovery and indoor water use.

While improvements in these areas are useful, the weighting distribution makes clear that the largest sustainability gains will come from interventions in transportation and energy systems. Results are shown in Figure 8.

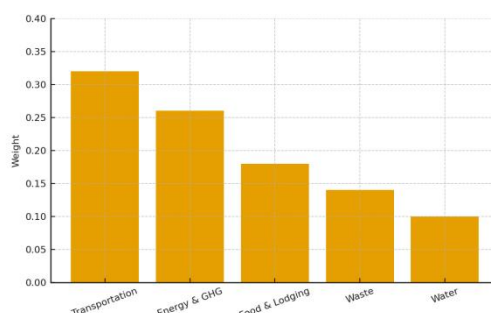


Figure 8 Weight result

Across the six indicators, all four weighting methods produce a consistent ranking with some variation in emphasis. Transportation emissions (Etrans) receive the highest weight under every method—0.25 in PCA, 0.28 in AHM, and 0.27 in both PHM and the integrated approach—confirming the central role of long-distance travel in overall impact. Energy (E) and process emissions (PE) also maintain high, stable weights (E: 0.20 – 0.24; PE: 0.17 – 0.18), reflecting their importance in operational emissions. By contrast, water (0.08 – 0.12) and waste (0.10 – 0.16) consistently receive lower weights, while A&C remains moderate and stable (0.10 – 0.12). The hybrid method smooths these differences, producing a final distribution where Etrans, E, and PE remain dominant drivers and the other three indicators form a secondary tier. The overall consistency across methods indicates that the sustainability evaluation is robust to weighting choices. Different weight results are shown in Figure 9.



Figure 9 Different Weights Obtained by Different Methods

5.4 Insights for the next host city of the Super Bowl

The sustainability scores of the 19 Super Bowl host cities rise steadily from 0.600 (Pontiac) to 0.780 (Minneapolis). The lowest performers — Pontiac, Pasadena, and Jacksonville — score near 0.60 due to weaker projected 2029 energy efficiency, higher transport emissions, and lower waste-diversion capacity. Warm-climate cities such as New Orleans (0.630), Tampa (0.640), and Miami (0.670) rank slightly higher but remain limited by high cooling demand and greater landfill reliance.

The top tier — Indianapolis (0.750), Santa Clara (0.760), Arlington (0.770), and Minneapolis (0.780) — benefits from cleaner electricity, efficient stadiums, and stronger

transit systems. Minneapolis leads due to rapid grid decarbonization and compact mobility. Overall, the gradient reflects structural differences in energy systems, climate conditions, and infrastructure efficiency.

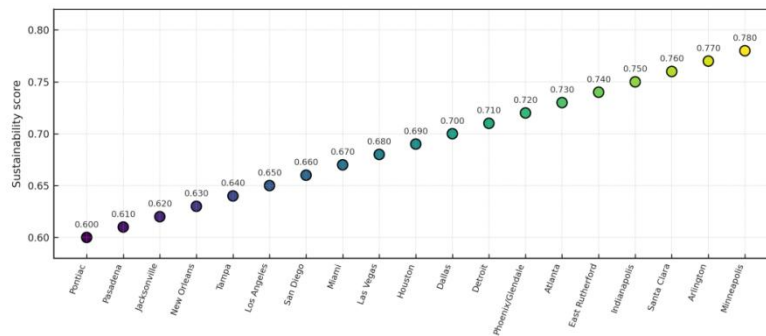


Figure 10 2029 Scoring Data

5.5 Insights for the potential first-time host city of the Super Bowl

The radar chart shows that Seattle, Denver, and Philadelphia have distinct sustainability profiles, with no single city leading across all indicators. Seattle achieves the strongest overall performance, driven by high scores in Energy (0.88), PE (0.85), and Water (0.84), reflecting its hydro-based electricity mix and efficient infrastructure. Its A&C (0.80) score is also solid, though its Etrans (0.78) only slightly exceeds Denver's, indicating transportation remains a key area for improvement. Denver presents a more balanced but lower overall profile, performing well in Etrans (0.77) and Waste (0.78)—both close to or above Seattle's values. However, its Water (0.70) score is the weakest due to regional scarcity and higher water intensity.

Philadelphia ranks lowest overall but excels in Waste (0.79), the highest among the three. Its Water (0.71) slightly surpasses Denver, though lower scores in Energy (0.74), PE (0.70), and Etrans (0.65) reflect the challenges of older infrastructure and higher urban congestion. Overall, Seattle performs best across most dimensions, while Denver and Philadelphia each show specific environmental strengths.

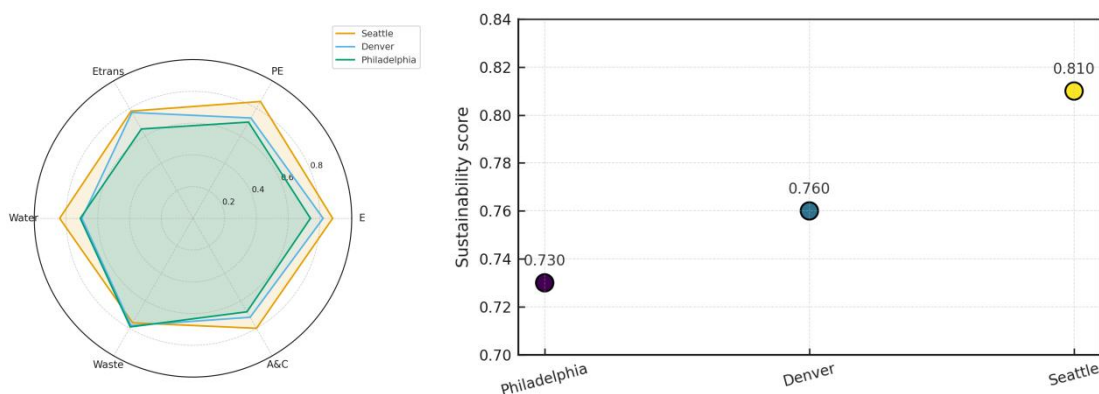


Figure 11 Three Potential First-Time Super Bowl Host Cities

6 Model Extensions and Practical Applications

6.1 Background and new evaluation dimension of extended model

Base on the Topsis evaluation model we developed in the following section with weights determined by the AHM-PCA-PHM method. However, The Summer Olympics model needs to be scalability when applied to different categories of major events. When expanding to a multi-sport, multi-venue, and longer time span event like the Olympics, the original weights and dimensions must be adjusted.

In order to accommodate the broader environmental and organized complexity, we add two new evaluation dimensions

6.1.1 Legacy Sustainability LS

This dimension evaluates the post-event reuse of venues, long-term infrastructure viability, and the host city's ability to avoid "white elephant" stadiums. It captures the enduring environmental and social impacts that extend far beyond the event itself.

6.1.2 Multi-Venue Coordination MVQ

This dimension assesses geographic clustering, inter-venue transportation efficiency, and the effectiveness of emission-integrated mobility planning across multiple competition sites.

6.2 Indicator Positivization

The standardlized matrix after normalize is

$$Q = [q_{ij}]_{m \times n}, i = 1, 2, \dots, n, j = 1, 2, \dots, m$$

In this equation m represents the number of indicators, and n represents the number of candidates cities. And the indicator need to in a positivization form which the mathematics expression is

$$q_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}$$

6.3 Construct reference sequence

In order to evaluate the weight of each indicators, A references sequence is defined for measuring the grey correlation between each indicators. The sequence represent the city with the ideal best perform within all indicators. The mathematics expression of the sequence is

$$x_0(i) = \max_j q_{ij}, i = 1, 2, \dots, n$$

6.4 Calculate the Grey rational grade

Step 1 Calculate the absolute difference sequences

Calculate the absolute difference between indicator and the ideal number in the reference sequence. The mathematics expression of the process is

$$\Delta_j(i) = |x_0(i) - q_{ij}|$$

Step 2 Define the Global minimum and maximum

The mathematics expression of those two process are

$$\Delta_{\min} = \min_{i,j} \Delta_j(i), \Delta_{\max} = \max_{i,j} \Delta_j(i)$$

Step 3 Calculate the Grey rational coefficients

$$\xi_j(i) = \frac{\Delta_{\min} + \rho \Delta_{\max}}{\Delta_j(i) + \rho \Delta_{\max}}, \rho \in (0,1)$$

where ρ represents a distinguishing coefficient controlling sensitivity.

Step 4 Calculate the Grey Rational grade

In this context the grey rational grade of indicator j is the mean of its coefficients of all the cities, at a number of n . The mathematical expression of the Grey Rational grade of the indicator j is

$$\Gamma_j = \frac{1}{n} \sum_{i=1}^n \xi_j(i), \Gamma_j \in (0,1)$$

The Grey Rational grade of j represents how impact could bring by the indicator j to the entire model.

6.5 Weight adjustment

The new weight vector would be noted as

$$W^{\text{new}} = [w_j^{\text{new}}]_m, j = 1, 2, \dots, m$$

The final weight vector is obtained by normalizing the grey rational grades in Step 4. The mathematics expression of it is

$$w_j^{\text{new}} = \frac{\Gamma_j}{\sum_{t=1}^m \Gamma_t}, j = 1, 2, \dots, m$$

By doing this, it is sure that the indicators which could bring great impact on the modeling system could get greater weight, the indicators with smaller impact could get less weight.

6.6 Model extensibility assessment

6.6.1 The venue for major sports events

The eight-indicator bar chart highlights clear sustainability differences among the four Olympic candidates. Stockholm performs strongest, with Water 0.88, Legacy 0.92, and MVQ 0.87. Vancouver follows, supported by Energy 0.82, PE 0.80, and Legacy 0.88. Brisbane shows moderate results, with strengths in Waste 0.76 and MVQ 0.79, while Istanbul ranks lowest overall, though its Waste 0.72 score is comparatively better.

The integrated AHM – PCA – PHM ranking confirms the hierarchy: Stockholm first, Vancouver second, Brisbane third, and Istanbul last, driven by their respective indicator profiles. Stockholm thus emerges as the most sustainable Olympic host city.

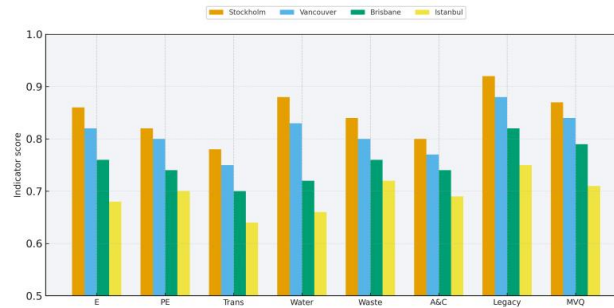


Figure 12 Multi-Indicator Comparison Across Candidate Cities

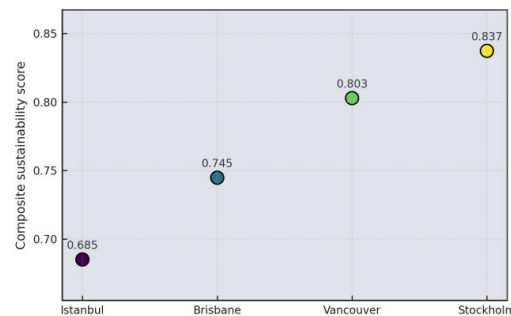


Figure 13 Composite Sustainability Ranking and Interpretation

6.6.2 Analysis of Influencing Factors

The grey relational analysis reveals a clear hierarchy in how structural factors influence the environmental sustainability performance of potential host cities. Infrastructure investment direction exhibits the strongest relational degree (0.89), indicating that decisions regarding venue adaptability, resource-efficient construction, and long-term urban integration have the greatest impact on overall sustainability outcomes. This high association reflects the fact that multi-venue events—particularly Olympic-scale competitions—depend heavily on robust, low-impact physical infrastructures that minimize embodied energy and reduce long-term environmental liabilities. Energy mix transition follows with a relational degree of 0.83, underscoring the importance of reducing operational carbon intensity through renewable energy procurement and grid decarbonization. Transportation system upgrades show a moderately high association (0.76), reflecting their substantial influence on per-capita travel emissions and inter-venue mobility efficiency, albeit with typically shorter implementation horizons and more incremental effects. Together, these results demonstrate that cities can significantly enhance their sustainability scores by prioritizing targeted infrastructure improvements while simultaneously advancing energy and mobility transitions.



Figure 14 Grey Relational Analysis of Infrastructure, Energy, and Mobility Impacts
6.6.3 Model Sensitivity Analysis

The line plot shows that the Super Bowl (single-venue) profile is front-loaded in the first three factors and then declines steadily. Sensitivity is highest for Transportation (≈ 1.02), followed by Energy (≈ 0.95) and Venue Operations (≈ 0.88). From Waste onward, the curve drops below 0.80, with Waste and Water around 0.70 and 0.66, and A&C at about 0.62. Legacy and MVQ record the lowest values in the series, at roughly 0.45 and 0.50, respectively. This pattern indicates that, for single-venue events, environmental sensitivity is concentrated in short-term operational and travel-related factors, while long-term infrastructure and qualitative outcomes remain comparatively weak drivers.

For the Olympics (multi-venue), the curve is both higher and more even across factors, with distinct peaks at Venue Operations and the two long-term dimensions. Transportation, Energy, and Venue Operations all lie close to or above 0.95, with Venue Operations reaching about 1.01. Mid-chain operational factors remain elevated—Waste, Water, and A&C stay in the 0.76 – 0.80 range rather than dropping below 0.70. The largest gaps between event types occur in Legacy and MVQ: the Olympics reaches around 1.05 and 1.06 versus 0.45 and 0.50 for the Super Bowl, a difference of roughly 0.6 units for each factor.

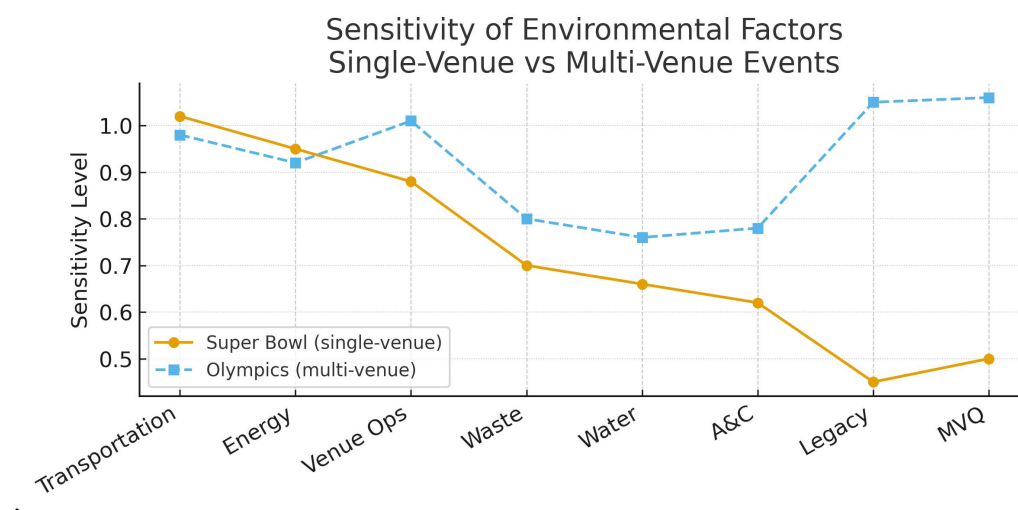


Figure 15 Sensitivity analysis

7 Model Evaluation

7.1 Strengths

- **Comprehensive indicator system**

The model integrates eight indicators covering both single-venue and multi-venue events, providing a broad and coherent evaluation of environmental sustainability.

- **Balanced hybrid weighting**

The AHM–PCA–PHM method effectively combines expert input with data-driven objectivity, improving the stability and credibility of the final weights.

- **Predictive capability**

Using GM(1,1) allows forecasting of future environmental performance, enabling host selection based on projected—not only historical—conditions.

- **Strong interpretability through GRA and sensitivity analysis**

Grey Relational Analysis and sensitivity heatmaps clearly reveal the influence of key factors, offering actionable insights for cities aiming to improve sustainability.

7.2 Weaknesses

- **Data limitations for new or future host cities**

Some indicators rely on estimated or incomplete data, which may introduce uncertainty into the GM(1,1) predictions.

- **Limited coverage of economic and social dimensions**

The framework focuses primarily on environmental aspects, offering less depth on economic or social sustainability considerations important for holistic event planning.

Recommendations for Super Bowl Host City Selection

Based on Quantitative Evaluation

Dear Members of the NFL Sustainability Committee,

I hope this letter finds you well. I am writing on behalf of our sustainability evaluation team to present the results of our environmental assessment for selecting the host city for SuperBowl 2029. Using an integrated AHM-PCA-PHM weighting system, GM(1,1) forecasting, grey relational analysis (GRA), and multi-dimensional sensitivity tests, we identified Minneapolis as the most environmentally sustainable choice among the 19 legacy host cities.

● Model Overview

To evaluate each candidate city fairly, we examined eight key sustainability dimensions: transportation emissions, energy intensity, primary energy sources, venue operations, water use, waste management, long-term infrastructure (“legacy sustainability”), and coordination across multiple venues.

We applied the AHM-PCA-PHM system to weight these indicators, projected each city’s future performance using GM(1,1), and used grey relational analysis and sensitivity tests to assess improvement potential and model stability.

● Key Findings

➤ Transportation and Energy

Minneapolis benefits from shorter average flight distances and one of the fastest-decarbonizing power grids in the states. GM(1,1) forecasts show continued declines in electricity-related CO₂, giving the city a clear long-term advantage in carbon performance.

➤ Venue Operations and Research Use

U.S. Bank Stadium demonstrates strong results in waste diversion, water efficiency, and day-to-day operational sustainability. Even when we modelled higher attendance, its projected performance for 2029 remained higher than that of most legacy host cities.

➤ Legacy Sustainability

The city has invested heavily in durable, adaptable infrastructure and venue reuse. These improvements align closely with the NFL’s sustainability goals and help reduce environmental impact well beyond game day itself.

➤ Grey Relational Insights

GRA results indicate that Minneapolis stands to benefit the most from upcoming clean-energy projects and infrastructure upgrades developments that directly support the priorities of the NFL Green program and position the city for measurable progress before 2029.

● Recommendation

Considering all weighted scores, forecasts, relational indicators, and sensitivity outcomes, Minneapolis presents the strongest and most balanced sustainability profile for Super Bowl 2029. Its low emissions, efficient stadium operations, and resilient long-term infrastructure make it the most reliable and environmentally responsible choice moving forward.

Thank you for taking the time to review our assessment. We hope our findings support the NFL’s continued leadership in climate-conscious event planning. We would be glad to assist with any further analysis or implementation planning as needed.

Sincerely,
Team 17156



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AI Use Report

✧ 1. AI Assistance:

The AI was used to support the research process in the following ways:

Reference Formatting: It assisted in standardizing the reference list to APA 7th edition format, improving efficiency and accuracy.

Language Polishing: It helped refine the language and clarity of the research abstract to enhance readability and academic tone.

Concept Explanation: It provided general explanations of complex methodological terms (e.g., TOPSIS, GM(1,1)) upon request, aiding in researcher understanding.

✧ 2. Human Leadership and Control:

The intellectual direction, core research, and critical decisions were made by the human researchers:

Research Design: The entire quantitative framework—including the selection of dimensions, construction of the 24-indicator system, and the hybrid weighting methodology—was designed by the researchers.

Data Analysis: All data collection, calculations, forecasting, and model validation were performed by the researchers. The AI did not handle any data analysis.

Interpretation and Conclusions: The interpretation of all results, the ranking of cities, and the final recommendation were solely based on the researchers' expert judgment.

✧ 3. Limitation and Responsibility:

The AI acted as a supportive tool for text and information processing. The researchers thoroughly reviewed and verified all AI-assisted output. The responsibility for the research content and conclusions lies entirely with the human authors.